

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement provides the operator proofs, verification conditions, transition construction, additional numerical results, and replication instructions for the fourth revision. It distinguishes continuous-time economic objects from finite transition approximations and distinguishes trained neural policies from analytical and finite-state regression tests. The main paper and this supplement use the same policy-evaluation, boundary, and temporal-self conventions.

S.1. OPERATOR PROOFS AND QUANTITATIVE APPROXIMATION

S.1.1. *Exact Policy Improvement*

We use the assumptions of the main paper’s exact separated-operator proposition. In particular, $I(p)(t, s)$ is a selector maximizing $\mathcal{H}^a[E(p)](t, s)$, not a stationary point of a parameterized actor objective. Continuity of E is in the norm controlling the value, time derivative, first spatial derivatives, second spatial derivatives, and the stated boundary traces. The admissible compact policy class is closed under I . These are assumptions on an exact boundary-value problem, not universal properties of neural policy classes.

PROOF OF EXACT SEPARATED-OPERATOR CONSISTENCY: Write $V_n = E(p_n)$ and $p_{n+1} = I(p_n)$. The evaluation equation gives $V_{n,t} + \mathcal{H}^{p_n}[V_n] = 0$. Pointwise maximization implies

$$V_{n,t} + \mathcal{H}^{p_{n+1}}[V_n] \geq 0.$$

The assumed comparison and verification inequalities, with the same boundary operator for both policies, imply $V_{n+1} \geq V_n$. Precompactness of the evaluation image implies boundedness and hence a pointwise limit $\bar{V}$. Every subsequential limit in the declared strong topology has this same value component. Its derivative components are the derivatives of that value, because uniform convergence of the functions and their derivatives preserves differentiation. Consequently all strong accumulation points coincide and $V_n \rightarrow \bar{V}$ strongly.

Let $p_{n_j} \rightarrow p$ be a policy subsequence, which exists by compactness. Continuity of evaluation gives $E(p) = \bar{V}$. Continuity of improvement at p and continuity of evaluation give

$$E(I(p)) = \lim_j E(I(p_{n_j})) = \lim_j V_{n_j+1} = \bar{V}.$$

Evaluating $I(p)$ at this common value yields

$$0 = \bar{V}_t + \mathcal{H}^{I(p)}[\bar{V}] = \bar{V}_t + \sup_a \mathcal{H}^a[\bar{V}].$$

The boundary traces also pass to the limit. Comparison identifies $\bar{V}$ with the optimal value, and verification identifies p as optimal because $E(p) = \bar{V}$. This argument does

not require $p = I(p)$ or convergence of the entire policy sequence. In the presence of tied maximizing actions, different optimal policies may share the same limiting value. *Q.E.D.*

The proof is an application of exact policy-improvement reasoning; it is not a claim of priority over [Jacka and Mijatović \(2015\)](#). The assumption of a compact strongly regular evaluation image can fail at corners or under degenerate diffusions. The stopped application is therefore evaluated by its specified positive transition approximation rather than asserted to satisfy every classical hypothesis globally.

S.1.2. Finite-Horizon Value-Loss Bound

For clarity, put $\Gamma_{n,n} = 1$ and $\Gamma_{n,j} = \prod_{\ell=n}^{j-1} \gamma_\ell$. Both $\mathcal{T}_n^a$ and its policy version are monotone and γ_n -Lipschitz in the sup norm. The implemented operator is $\tilde{\mathcal{T}}_n^a$. The three nonnegative error bounds at $\hat{v}_{n+1}$ are

$$\begin{aligned}\epsilon_n &\geq \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \delta_n &\geq \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \kappa_n &\geq \sup_a \left\| (\tilde{\mathcal{T}}_n^a - \mathcal{T}_n^a) \hat{v}_{n+1} \right\|_\infty.\end{aligned}$$

Here $\tilde{\mathcal{T}}_n = \sup_a \tilde{\mathcal{T}}_n^a$, and $\|\hat{v}_N - G\|_\infty \leq \eta$.

PROOF OF THE MAIN VALUE-LOSS THEOREM: Let $D_n = \left\| \hat{v}_n - J_n^{\hat{p}} \right\|_\infty$. Inserting the implemented and target fixed-policy backups gives

$$D_n \leq \epsilon_n + \kappa_n + \gamma_n D_{n+1}.$$

Let $R_n = \|\hat{v}_n - V_n^*\|_\infty$. The supremum inequality $|\sup_a f_a - \sup_a g_a| \leq \sup_a |f_a - g_a|$ and the nonnegative improvement gap give

$$R_n \leq \epsilon_n + \delta_n + \kappa_n + \gamma_n R_{n+1}.$$

Backward iteration, with $D_N, R_N \leq \eta$, proves

$$\begin{aligned}D_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta, \\ R_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \delta_j + \kappa_j) + \Gamma_{n,N} \eta.\end{aligned}$$

Feasibility implies $V_n^* \geq J_n^{\hat{p}}$. Combining the two inequalities proves the stated upper bound with coefficients $2\epsilon_j + \delta_j + 2\kappa_j$ and terminal coefficient 2η . *Q.E.D.*

The theorem is finite-horizon. It permits a recursive operator with $\gamma_n > 1$ but then records amplification explicitly through $\Gamma_{n,j}$. It does not imply contraction for an arbitrary Epstein–Zin aggregator. Boundary killing can be included in the transition kernel by treating the boundary payoff as a known reward component; only continuation mass contributes to γ_n .

S.1.3. Differential Residual and Boundary Error

Let v be a $C^{1,2}$ candidate for the additive, stopped, discounted problem. Define $R^a = v_t + \ell^a + \mathcal{L}^a v - \rho v$. Suppose $|R^\beta| \leq \epsilon(t)$, $\sup_a R^a - R^\beta \leq \delta(t)$, and the payoff discrepancy at terminal time or exit is at most η . Assume the localized Itô formula passes to expectations by uniform integrability.

For every admissible p , stopped Itô integration gives

$$J^p(t, s) - v(t, s) = \mathbb{E}_{t,s}^p \left[\int_t^\tau e^{-\rho(r-t)} R^{p_r}(r, S_r) dr + e^{-\rho(\tau-t)} (G - v)(\tau, S_\tau) \right].$$

The upper bound for an arbitrary policy is $\eta + \int_t^T e^{-\rho(r-t)} (\epsilon(r) + \delta(r)) dr$. The absolute discrepancy for $\hat{p}$ is at most $\eta + \int_t^T e^{-\rho(r-t)} \epsilon(r) dr$. Taking a supremum over p and adding proves the differential corollary. The argument is conditional on uniform residual and boundary bounds. Replacing these bounds by sampled root mean squared losses is not a valid proof step.

S.1.4. Feasible Candidates, Convex Gaps, and Reference Bounds

If $Q(s, a)$ is L_a -Lipschitz in a and a feasible set $\mathcal{C}(s)$ is an h_a -net of the admissible actions, then a maximizer over $\mathcal{C}(s)$ is within $L_a h_a$ of the full supremum. Adding a neural proposal to $\mathcal{C}(s)$ cannot reduce its achieved backup value. Neither statement bounds the evaluation error of the critic that generated Q .

For differentiable convex F on a compact convex set A , its tangent inequality implies

$$F(a) - \inf_{b \in A} F(b) \leq \nabla F(a)^\top a - \min_{b \in A} \nabla F(a)^\top b =: g_F(a).$$

For $A = \{a \geq 0 : \sum_i a_i \leq B\}$, the minimum is $B \min(0, \min_i \partial_i F(a))$. The same formula applies to a scenario tree with separate feasible actions at each information node: sum the node-specific linear minimization gaps, using the gradient of the complete probability-weighted objective. This gradient must include all descendant costs.

If a feasible reference has objective C_R and gap G_R , then $C_R - G_R \leq C^* \leq C_R$. Any other feasible policy, evaluated on the same tree, therefore satisfies

$$C_\pi - C_R \leq C_\pi - C^* \leq C_\pi - C_R + G_R.$$

The program evaluates both objectives by all 64 terminal branches. It does not insert the neural critic as a substitute for policy cost. The reported interval is based on ordinary double-precision evaluation of these inequalities; it is not an interval-arithmetic enclosure of rounding error.

A useful conditional policy bound also follows. If the optimal continuation backup $Q^*(s, \cdot)$ is μ -strongly concave and the approximate continuation is within e of the optimal one, an action whose approximate improvement gap is at most δ obeys

$$\|\hat{a} - a^*\|^2 \leq \frac{2}{\mu} (\delta + 2\gamma e).$$

Indeed the value of each action changes by at most γe , and strong concavity bounds the resulting true suboptimality below by $\mu \|\hat{a} - a^*\|^2 / 2$. This is not applied to a nonconcave NDU maximand. In the convex resource problem the immediate quadratic cost

supplies strong convexity $0.2/d$ for minimization. Its deterioration with dimension is included in the conditioning discussion rather than hidden in a dimension-free speed claim.

S.2. COUNTEREXAMPLES, TRACE ESTIMATION, AND DIFFERENTIATION

S.2.1. *Why a Joint Penalty Does Not Define the Economic Solution*

Consider $-V' + V = 0$ on $[0, 1]$, $V(1) = 1$. Its solution is $V^*(t) = e^{t-1}$. The historical composite objective was of the form

$$L(V) = \int_0^1 \{V(t) + [-V'(t) + V(t)]^2\} dt.$$

The feasible perturbation $V_\varepsilon(t) = V^*(t) - \varepsilon(1 - t)$ preserves the terminal condition but has residual $-\varepsilon(2 - t)$. Consequently

$$L(V_\varepsilon) - L(V^*) = -\frac{\varepsilon}{2} + \frac{7\varepsilon^2}{3} < 0 \quad \text{for } 0 < \varepsilon < \frac{3}{14}.$$

Thus the exact economic solution is not even a local minimizer of this penalty. Separate parameter blocks, objectives, and verification conditions are substantive changes, not a change of notation for that joint loss.

S.2.2. *A Stable Actor Stationary Point Need Not Be Optimal*

On $[-2, 2]$, let $H(a) = -(a^2 - 1)^2 + 0.2a$. Its stationary equation is $4a^3 - 4a - 0.2 = 0$. There are three real roots. At the negative local maximizer, approximately -0.9739943532 , $H''(a) = 4 - 12a^2 < 0$, so gradient ascent has a locally asymptotically stable equilibrium. The positive maximizer is approximately 1.0241203002 and has higher payoff by approximately 0.3998745872 . Both are isolated. The example embeds into a finite-horizon control problem with flow $H(a)$ and exact evaluation $(T - t)H(a)$.

Consequently, even exact fast evaluation, diminishing separated step sizes, bounded iterates, and zero noise do not turn isolated actor stationary points into policy-improvement fixed points. Under a suitable stochastic-approximation theorem the justified conclusion concerns the limiting invariant or stationary set. Global optimality additionally requires an argument excluding suboptimal attractors. Convex improvement in the resource example supplies such an argument for each surrogate action problem; it does not automatically eliminate critic approximation error. Constant-step Adam has no convergence theorem in this paper, and the executed neural runs use the explicitly recorded L-BFGS and constrained least-squares routines instead.

S.2.3. *Small Strong Residuals Can Miss a Viscosity Solution*

For the exit-time problem $dX_t = a_t dt$, $|a_t| \leq 1$, with flow -1 and zero payoff on $\{-1, 1\}$, the value is $V(x) = |x| - 1$. The viscosity equation is $1 - |V'| = 0$. The smooth functions

$$W_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2}$$

satisfy the boundary data, while their squared strong residual under the uniform distribution tends to zero:

$$\frac{1}{2} \int_{-1}^1 \left(1 - \frac{|x|}{\sqrt{x^2 + \varepsilon^2}}\right)^2 dx \sim \left(2 - \frac{\pi}{2}\right) \varepsilon.$$

Nevertheless $W_\varepsilon \rightarrow 1 - |x|$, which has the wrong sign and is not the desired viscosity solution. The asymptotic expression follows by $x = \varepsilon y$ and integration of the resulting integrable function on $[0, \infty)$. The code also evaluates the finite- ε integral by independent adaptive quadrature.

Monotonicity, stability, and consistency with an appropriate comparison principle are central to viscosity approximation; see [Barles and Souganidis \(1991\)](#). Smooth network approximation of a function and its derivatives, as in [Hornik, Stinchcombe, and White \(1990\)](#), concerns a smooth target and does not approximate a derivative at a kink or replace a viscosity-selection argument. Historical kink and residual arrays remain available as archival material, not as evidence that an unregularized strong-residual loss selects the correct solution.

S.2.4. Probe Budget and the Squared-Loss Estimand

Let $A = B^\top \nabla^2 v B$ be symmetric, z_k independent standard Gaussian vectors, and $q_k = z_k^\top A z_k$. Gaussian fourth moments give $\mathbb{E} q_k = \text{tr } A$ and $\text{Var}(q_k) = 2 \|A\|_F^2$. For deterministic a and $\bar{q}_K = K^{-1} \sum_k q_k$,

$$\mathbb{E} \left(a - \frac{1}{2} \bar{q}_K \right)^2 = \left(a - \frac{1}{2} \text{tr } A \right)^2 + \frac{\|A\|_F^2}{2K}.$$

In contrast $K^{-1} \sum_k (a - q_k/2)^2$ has excess expectation $\|A\|_F^2/2$, independent of K . For the regression matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\|A\|_F^2 = 15$, $a = 1.25$, and the exact-trace loss is 1.5625. The population excess is $7.5/K$.

An unbiased alternative for $K \geq 2$ is the off-diagonal statistic

$$U_K = \frac{(\sum_k r_k)^2 - \sum_k r_k^2}{K(K-1)}, \quad r_k = a - q_k/2.$$

Independence implies $\mathbb{E} U_K = (\mathbb{E} r_k)^2$. Under an integrable derivative envelope, differentiating its expectation yields the gradient of the exact squared residual. The realized statistic can be negative; unbiasedness neither makes it a nonnegative sample loss nor guarantees optimizer stability. If the action or parameters are selected using the same probes, the fixed-parameter identity is not an adaptive generalization theorem. Independent held-out probes are needed to interpret such selection.

Each row averages 20,000 independent batches. The Monte Carlo standard error pertains to the repeated-batch mean, not to a single quadratic form or its within-batch variance. The numerical test checks the known expectation within six Monte Carlo standard errors and preserves the wrong mean-of-squares statistic as a negative control. It does not assign value or policy errors to this algebraic experiment.

TABLE S.1
INDEPENDENT-BATCH STOCHASTIC-TRACE LOSSES

K	Correct mean	Population	MC s.e.	Old mean	Debiased mean
1	8.91810	9.06250	0.22234	8.91810	–
4	3.47199	3.43750	0.04758	9.17095	1.57233
16	2.03146	2.03125	0.01492	9.06287	1.56270
64	1.68023	1.67969	0.00640	9.06132	1.56307

20,000 independent Gaussian batches per row; the exact-trace squared residual is 1.5625. Correct mean averages probes before squaring; old mean averages individual squared residuals. The debiased off-diagonal statistic requires at least two probes. Its standard errors and both variance estimands are retained in the JSON record.

S.2.5. Executable Computational-Graph Tests

The test suite forms a quadratic critic and a control-dependent diffusion loading, contracts its automatic Hessian with the covariance, and checks both parameter blocks. In the actor update, critic coefficients receive no gradient but the action-dependent covariance does. In the critic update, the policy output is detached while the critic coefficients receive gradients. A finite-transition actor derivative is also checked against central differences in all three controls. Freezing critic parameters does not detach the next-state input: the continuation gradient with respect to the action-induced state is retained.

A separate test compares the resource critic’s explicit gradient with PyTorch automatic differentiation. The convex squared-ReLU feature architecture is a transparent special case of the input-convex approach of [Amos, Xu, and Kolter \(2017\)](#). Exact directional second derivatives or Hessian-vector products, following [Pearlmutter \(1994\)](#), can avoid constructing a dense Hessian when the diffusion has low rank. The complete costs still depend on network size, diffusion multiplication, sample count, probe count, and optimization iterations.

S.3. THE STOPPED PREFERENCE-FORMATION ECONOMY

S.3.1. Admissibility and the Boundary Contract

The interior state is $(u, X) \in (1.2, 2.8) \times (0.5, 2)$, with the controls and coefficients stated in the main paper. The process is stopped at the first exit or $T = 1$, whichever occurs first. It receives $G(u, X) = -0.02(u - 2)^2 + 0.1 \log X$ at that time. A start on the boundary is settled immediately. This is a finite-horizon contract on an open operating domain; it is not a claim that the unstopped diffusion remains in a rectangle.

For bounded progressively measurable controls, the wealth SDE with coefficients linear in wealth and the additive preference SDE have a well-defined solution up to exit. Before exit, wealth and consumption are bounded away from zero. The flow, terminal payoff, and quadratic adjustment cost are bounded on the stopped domain. Thus the additive objective is finite without introducing a reflected local time or a wealth-transfer process.

The original state-constraint interpretation cannot hold with these coefficients. Preference volatility is 0.05 at both endpoints, and the maximal wealth drift at the lower boundary is $(0.02 + 0.8 \times 0.06)0.5 - 0.05 = -0.016$. At zero portfolio volatility it is more negative. Neither fact is repaired by projecting an Euler endpoint back onto the

rectangle. The stopped contract retains all interior coefficients and makes the change in boundary economics explicit.

S.3.2. A Boundary-Compatible Payoff

The choice of G has a useful analytic check. Its derivatives are $G_u = -0.04(u - 2)$, $G_{uu} = -0.04$, $G_X = 0.1/X$, $G_{XX} = -0.1/X^2$, and $G_{uX} = 0$. For every admissible action and every $k \geq 0$,

$$\begin{aligned} \ell^a + \mathcal{L}^a G - \rho G &\leq \max_{u \in [1, 2.8]} \frac{0.8^{1-u}}{1-u} + 0.0064 + 0.0043 - 0.00005 + 0.0032846 \\ &< -0.8162. \end{aligned}$$

The consumption bound follows because utility is increasing in consumption. Maximization over u at $c = 0.8$ gives the upper endpoint; it can also be checked by the explicit derivative. The 0.0043 term bounds $0.1[0.02 + 0.06\pi - c/X]$ by using $\pi \leq 0.8$, $c/X \geq 0.025$. The omitted portfolio diffusion term is nonpositive, and so are adjustment costs. The last term bounds $-\rho G$ uniformly.

Stopped Itô integration therefore implies $J^p \leq G$ for every admissible policy. To obtain boundary continuity from below, fix for example a constant admissible nonzero portfolio share. Its two-dimensional covariance is uniformly positive definite on the closed rectangle because $X \geq 0.5$ and $|\varrho| < 1$. The boundary is regular for this fixed diffusion, and its expected exit time tends to zero as the start tends to a boundary point. Bounded rewards and continuity of G imply that this policy's stopped payoff tends to G there. Since its payoff is bounded above by the optimal value and the latter is bounded above by G , the value has the declared boundary limit. This argument establishes boundary compatibility; it does not assert global classical differentiability at corners.

S.3.3. Transition, Covariance, and First Exit

For step h , form four equiprobable sign pairs $(z_1, z_2) \in \{-1, 1\}^2$. Let

$$\begin{aligned} d_u &= \theta h + \sigma_u \sqrt{h} z_1, \\ d_X &= [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} [\varrho z_1 + \sqrt{1 - \varrho^2} z_2]. \end{aligned}$$

For an interior state s , α is the first fraction in $[0, 1]$ at which the line $s + \alpha d$ meets a face, or one if it stays inside. An exit branch has payoff

$$\frac{1 - e^{-\rho \alpha h}}{\rho} \ell(s, a) + e^{-\rho \alpha h} G(s + \alpha d).$$

A surviving branch instead uses the continuation value at $s + d$ and the full-step flow annuity. At a boundary start the backup is exactly $G(s)$. In particular there is no nonzero continuation from a boundary node. The reference evaluator uses nonnegative bilinear interpolation weights that sum to one on surviving branches.

These signs reproduce conditional first moments and the covariance matrix of the written diffusion. Polynomial tests on $1, u, X, u^2, X^2, uX$ distinguish covariance terms

from drift-product terms of order h . At the test state and action the covariance is

$$\begin{pmatrix} 0.0025 & -0.002 \\ -0.002 & 0.0256 \end{pmatrix}.$$

The largest generator discrepancy decreases by a factor of ten when h decreases from 10^{-2} to 10^{-3} and again to 10^{-4} . Preference controls of -0.2 and 0.2 give different mean next states even when both displacements are smaller than a spatial grid cell. No nearest-neighbor map erases the control.

The line-exit rule is an approximation to diffusion killing, not exact Brownian bridge detection. Before-correction overshoots, branch-exit frequencies, occupied exit probabilities, and post-exit payoffs are distinct recorded objects. A reflection-regulator field is inapplicable and is represented by null with that explanation. For the joint state-time refinement, the squared largest mesh width divided by h decreases as $0.04, 0.02, 0.01$; the separate control refinements are also retained. These observations support consistency diagnostics but do not supply a numerical bound for every κ_n in the main theorem.

S.3.4. *Economic Comparative Statics*

For each admissible policy let $B(p)$ denote cost-free benefit. Define discounted cumulative effort by $C(p) = \mathbb{E} \int_0^{\tau_p} e^{-\rho t} \theta_t^2 / 2 dt$. The stopping time is policy dependent but not directly dependent on k for a fixed policy. Thus $V(k) = \sup_p [B(p) - kC(p)]$ is decreasing and convex. For optimizers p_1, p_2 at $k_1 < k_2$, revealed optimality gives

$$\begin{aligned} B(p_1) - k_1 C(p_1) &\geq B(p_2) - k_1 C(p_2), \\ B(p_2) - k_2 C(p_2) &\geq B(p_1) - k_2 C(p_1). \end{aligned}$$

Adding implies $(k_2 - k_1)[C(p_2) - C(p_1)] \leq 0$. Also $-C(p)$ is a subgradient of V at any k for which p is optimal, and $V'(k) = -C(p)$ wherever V is differentiable. For policies with value losses at most e_1, e_2 , the same inequalities with these errors imply

$$C(p_2) - C(p_1) \leq \frac{e_1 + e_2}{k_2 - k_1}.$$

The result applies to the continuous stopped model and separately to each exactly specified positive finite approximation. A finite-grid comparison cannot set $e_i = 0$ for the continuous problem merely because its discrete maximization was exhaustive.

The reported effort is computed both backward and by forwarding the induced state probabilities. The latter multiplies live branch mass by bilinear weights and accumulates discounted cost through the line-exit fraction. This is an occupation calculation for the interpolation-defined finite Markov chain. It provides an economically interpretable sum rather than an unweighted mean of control magnitudes. A reduction of this aggregate does not imply a pointwise reduction of θ , consumption, or portfolio share. Nor does a comparative static identify k from observations without a specified data model and an injective moment map.

S.4. PORTFOLIO AND RECURSIVE-UTILITY VERIFICATION

S.4.1. *Stationary Merton Benchmark*

Write $b = 1 - \gamma$, $V(X) = AX^b/b$, $c = mX$, and

$$g(m, \pi) = r + (\mu - r)\pi - m + \frac{1}{2}(b - 1)\sigma^2\pi^2.$$

For exponential utility discount ρ , the fixed-policy stationary equation is

$$A = \frac{m^b}{\rho - bg(m, \pi)}$$

on the positive branch $\rho - bg > 0$. Optimization gives $\pi^* = (\mu - r)/(\gamma\sigma^2)$ and

$$m^* = \frac{\rho - (1 - \gamma)[r + (\mu - r)^2/(2\gamma\sigma^2)]}{\gamma}.$$

At $(\rho, \gamma, \mu, r, \sigma) = (0.04, 2, 0.08, 0.02, 0.2)$ the targets are $(m^*, \pi^*) = (0.04125, 0.75)$. Wealth remains positive under this proportional policy. Its discounted terminal candidate expectation decays at rate $\rho - bg(m^*, \pi^*) = m^* > 0$, verifying its own transversality condition. The usual verification comparison is made over policies for which the stopped local martingale and discounted terminal inequalities pass to the limit.

The executed feature solver treats $q = \log A$, the consumption ratio, and the portfolio ratio as unknowns. It alternates the fixed-policy scalar evaluation root with an automatic-differentiation actor optimization, without feeding the analytical targets into the updates. Positivity is imposed by $A = e^q$. This is a learned homothetic feature model; the ratios alone would not be a neural-function benchmark on an unrestricted state space. The no-short regression at $\mu = 0.01$ instead has unconstrained share -0.125 and constrained share zero at every positive wealth. It is not a wealth-dependent switching problem.

S.4.2. *Epstein–Zin Driver and the Admissible Comparison Class*

Use $a = 1 - 1/\psi$, $b = 1 - \gamma$, with $a, b \neq 0$, $c > 0$ and $bV > 0$. The driver is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}.$$

The log limits must be taken separately when a denominator vanishes. For a homothetic candidate $V = AX^b/b$, dividing the stationary HJB by AX^b gives

$$\frac{\rho}{a} \{m^a A^{-a/b} - 1\} + g(m, \pi) = 0.$$

Thus a fixed proportional policy has a unique positive homothetic coefficient whenever $1 - ag/\rho > 0$. Its consumption first-order condition is $\rho m^{a-1} A^{-a/b} = 1$, and the portfolio first-order condition gives the same Merton share with risk aversion γ . Eliminating A yields

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2},$$

$$m^* = \psi\rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma\sigma^2} \right],$$

$$A^* = (\rho(m^*)^{a-1})^{b/a}.$$

For $(\rho, \gamma, \psi, \mu, r, \sigma) = (0.04, 5, 1.5, 0.08, 0.02, 0.2)$ this gives $m^* = 0.0455$ and $\pi^* = 0.3$. The normalized Hamiltonian is strictly concave in the consumption ratio and portfolio share on the specified positive domain. The selected ratios lie inside the implemented action bounds.

For precision about infinite horizon, the verification class consists of policies for which the policy-wise recursive equation exists on the branch $bY > 0$, its localized comparison argument is integrable, and the comparison-weighted terminal remainder vanishes. On compact subsets of $c > 0, bY > 0$ the driver is locally Lipschitz, but this does not imply a global Lipschitz constant on the entire half-line. One may first stop at compact localization sets, apply the mean-value linearization in the value argument, and then impose uniform integrability and the weighted terminal limit to remove localization. With $\Gamma_t = \exp(\int_0^t f_V(c_s, \zeta_s) ds)$ for the intermediate value ζ_s , the required remainder is of the form $\mathbb{E}[\Gamma_T |V(X_T) - Y_T^p|] \rightarrow 0$. The candidate HJB inequality and this comparison yield $Y_0^p \leq V(X_0)$, with equality for the maximizing policy.

For the selected homothetic policy itself, $f_V = -0.1115$, while the growth exponent of $\mathbb{E}X_t^b$ is $bg = 0.066$. The candidate comparison-weighted value therefore decays at rate $0.1115 - 0.066 = 0.0455 > 0$. The numerical record evaluates the fixed-policy equation and target errors, and the analytic calculation verifies this decay. This specifies a genuine stationary recursive benchmark with an explicit admissible verification class; it does not assert that all bounded controls satisfy the terminal condition automatically.

The sign-enforcing architecture is actually executed: $V = e^q X^b / b$, hence $bV = e^q X^b > 0$. A separate finite-horizon architecture test uses

$$V_\xi(t, X) = \frac{1}{b} \exp\{\log[bG(X)] + (T - t)N_\xi(t, X)\}$$

for $bG > 0$. It verifies the terminal identity and sign on test points. It is a domain test, not evidence that an additional finite-horizon recursive problem was trained. These distinctions follow the policy-wise recursive-utility formulation of [Duffie and Epstein \(1992\)](#) rather than replacing recursion by a pointwise evaluation at $c = 1, V = -1$.

S.4.3. A Structured Stochastic Dynamic Control Check

For $d \in \{4, 8, 16\}$, the additional six-period LQR problem uses

$$A = 0.82I + 0.07(J + J^\top), \quad B = 0.2I, \quad Q = I + 0.31\mathbf{1}^\top/d, \quad R = 0.15I,$$

and Gaussian shock loading $\Sigma = 0.08(I + 0.31\mathbf{1}^\top/d)$, with terminal matrix Q . Here J has ones on its upper adjacent diagonal. The linear neural actor and quadratic feature critic are trained by policy evaluation and improvement. The independent reference uses the finite-horizon Riccati recursion, including the noise contribution to the value constant. A second recursion evaluates the learned fixed policy rather than using its fitted critic as a performance score.

The experiment has cross-state dynamics and non-diagonal stochastic covariance but no binding control constraints. It is retained as a structured dynamic regression test. The main shared-budget resource experiment supplies the separately executed constrained case; neither is relabeled as an adaptive sparse-grid comparison.

TABLE S.2
STRUCTURED STOCHASTIC LQR REGRESSION

d	Actor coeff. error	Cost excess max.	Neural sec.	Riccati sec.
4	4.25e-06	3.03e-11	0.0571	0.00037
8	2.25e-06	2.14e-11	0.0724	0.00028
16	4.30e-06	3.84e-11	0.1330	0.00033

Six periods, linear actors and quadratic feature critics, independent Riccati reference, and separate frozen-policy evaluation. Cost excess is normalized per dimension in the implementation. This structured unconstrained regression is not the main binding-resource experiment and supplies no generic neural speed advantage.

S.5. TEMPORAL SELVES AND DYNAMIC COMPETITION

S.5.1. *Sophisticated Consumption–Savings*

Let one period have length Δ , $\delta = e^{-\rho\Delta}$, and terminal payoff $G(w) = \log w$. The acting self chooses $c \in (0, w)$ to maximize $\Delta \log c + \beta \delta V_{n+1}(w - c)$, whereas ordinary continuation evaluates the chosen policy as

$$V_n(w) = \Delta \log c_n(w) + \delta V_{n+1}(w - c_n(w)).$$

If $V_{n+1} = A_{n+1} \log w + B_{n+1}$, strict concavity gives

$$\begin{aligned} z_n &:= \frac{c_n(w)}{w} = \frac{\Delta}{\Delta + \beta \delta A_{n+1}}, \\ A_n &= \Delta + \delta A_{n+1}, \\ B_n &= \Delta \log z_n + \delta A_{n+1} \log(1 - z_n) + \delta B_{n+1}. \end{aligned}$$

Starting with $A_N = 1, B_N = 0$ yields an exact sophisticated equilibrium. The coefficient A_n is independent of β , but the policy and the constant are not. The reference test uses $\Delta = 1$, four consumption dates, $\rho = 0.04$, and $\beta \in \{0.7, 1\}$. At four wealth levels it separately solves the bounded one-shot consumption problem and checks the evaluation identity with the frozen continuation coefficients.

For two consumption dates, $\delta = 1$, the correct first-date ratio is $1/(1 + 2\beta)$. Reusing the acting self's maximand as its stored continuation would instead yield $1/(1 + \beta + \beta^2)$. At $\beta = 0.7$ the difference is 0.4166667 versus 0.4566210. The $\beta = 1$ check alone cannot reveal this error. These checks implement the written temporal game rather than an ordinary geometric-discount control problem. They do not identify the finite-state reference as a trained neural solution.

S.5.2. *A Stochastic Capacity Duopoly*

There are four decision dates. Demand is $D \in \{1, 1.3\}$ with probability 0.8 of staying at its current value. Each firm's installed capacity $K_i \in \{0, 1\}$ determines current output $q_i = K_i/3$. Current operating profit is $q_i(D - q_i - q_j)$. Firm i chooses a maintenance or renewal intensity $a_i \in [0, 1]$ at cost $0.4a_i^2/2$. Independently across firms conditional on actions, next-period capacity is one with probability a_i and zero otherwise. The discount factor is 0.95 and terminal value is $0.05K_i$. There are eight Markov states and four dates, so complete state enumeration is possible.

For fixed rival intensity, the continuation value is affine in a firm's own intensity. Its best response is therefore the projection onto $[0, 1]$ of its discounted marginal continuation gain divided by 0.4. The joint map is affine before projection, with slope determined by the cross difference of the four next-capacity continuation values. The program checks the product of the two own best-response slopes, iterates from multiple initial profiles, and compares their limits. This finite example has a contraction bound below one. It does not establish uniqueness in arbitrary dynamic games.

An independent backward best-response recursion freezes the rival's entire state-contingent policy at every subsequent date and allows the tested firm to deviate at all dates. Its difference from the profile value is dynamic exploitability. This is stronger than checking a single current Hamiltonian against an inconsistent continuation. The report records the maximum over all dates, states, and players, with a numerical tolerance of 10^{-10} . A separate automatic-differentiation test confirms that a player's loss does not differentiate the rival's parameters.

The static special case $q_i(1 - q_i - q_j)$ remains a useful negative control. Symmetric Nash output is $1/3$, whereas joint-profit output is $1/4$. At the latter profile, changing one firm's quantity to its own best response is profitable. A symmetric or jointly optimal actor graph is therefore not a substitute for a Nash calculation. The dynamic capacity model is a finite-state equilibrium computation; no competitive limit or general neural-game convergence is inferred from it.

S.5.3. Market-Size Normalization and a Conditional Small-Player Bound

To retain the economically relevant many-firm extension without conflating different economies, consider per-firm normalized output y_i and actual output $q_i = y_i/N$, with inverse demand $P = a - b\bar{y}$, $\bar{y} = N^{-1} \sum_i y_i$. Multiplying each firm's monetary payoff by N gives normalized flow $y_i P - C(y_i)$ under consistently scaled costs. Its direct strategic price derivative is $-by_i/N$. Investment, depreciation, and state transitions must be scaled and retained as well; this current-flow derivative by itself does not establish convergence of dynamic equilibria.

A precise conditional implication is available. Suppose that, against a fixed rival profile, the finite- N player's transition operators and a price-taking comparison operator are monotone and γ_n -Lipschitz, have the same terminal payoff, and differ uniformly by at most η_n/N on all policies and continuation values used in the comparison. Then a policy optimal for the price-taking problem loses at most

$$\frac{2}{N} \sum_{j=n}^{N-1} \Gamma_{n,j} \eta_j$$

relative to the finite- N best response. To prove it, backward iteration bounds the difference of the two optimal values by $N^{-1} \sum_j \Gamma_{n,j} \eta_j$, and bounds the two evaluations of that fixed policy by the same amount. Their sum gives the result. Uniform Lipschitz control of aggregate-state effects is one route to the operator hypothesis, but it must be checked in a specified model. This proposition concerns a unilateral deviation bound, not selection or convergence of a sequence of equilibria. It is theoretical extension material and is not recorded as an executed large- N experiment.

S.6. REPLICATION AND ADDITIONAL RESULTS

S.6.1. *Source and Evidence Identities*

The authoritative sources are `ECTA_R4.tex`, `SUPP_R4.tex`, their explicitly included sections, and `replication/r4/`. The base review is pinned at commit 79a7d84. The replication manifest records the reachable source commit, full SHA-256 digests of the source files, package versions, and output identities. Source files are committed before their authoritative execution. A later result commit can add generated outputs without changing that source anchor.

Execution status, evidence class, and numerical-tolerance status are separate fields. A completed program is not automatically a solved control problem. The coupled resource runs have a specified held-out cost-loss target and independent reference gaps. The NDU comparison reports discrepancies and does not assign a passed uniform PDE certificate. Algebraic checks, domain tests, finite-state equilibria, and structured feature solvers each have their own evidence class. An unavailable continuous-time error bound or reflection regulator is null with an explanation, not a numerical zero.

Scientific digests exclude elapsed time and hardware measurements. File digests still identify complete files, including timing records. Floating-point algorithms, package versions, and BLAS implementations can change last digits across platforms. The source and saved weights permit direct policy replay independently of retraining. Timings include neural fitting and full evaluation for the stated initial-state batch; they do not imply a universal hardware comparison or unmeasured amortization advantage.

S.6.2. *Architectures and Optimization*

For the stopped preference model the safeguarded neural run uses separate two-hidden-layer tanh actors of width 32 and critics of width 64 at each of eight decision dates. The actor has three bounded output coordinates. The critic writes the value as G plus remaining time times a product vanishing on all spatial faces times its network output. The training sample contains 2,048 interior states. L-BFGS caps are 160 actor and 1,000 critic iterations with strong-Wolfe line search; actual function evaluations and iteration counts are recorded when produced by the optimizer. The implementation never trains on the independent grid reference’s values or policies.

The unsafeguarded pilot uses width-32 critics and 1,024 training states with a smaller critic budget. It is retained as a development baseline, not used as a matched causal ablation. The resource experiment supplies a matched ablation: the same fitted actor weights are evaluated with and without pointwise improvement at deployment. It holds training, initial states, shocks, and model primitives fixed.

For resources the convex critic is

$$q(x) + \ell^\top(x - \mathbf{1}) + c_0 + \sum_{j=1}^{6d+102} \alpha_j [\max\{w_j^\top(x - \mathbf{1}) + b_j, 0\}]^2, \quad \alpha_j \geq 0.$$

Directions consist of signed coordinate axes, signed aggregate directions, and 32 seeded Gaussian directions, each at offsets $-0.5, 0, 0.5$. The coefficients are fitted by bounded least squares with a 10^{-5} ridge term. The actor has one width-32 tanh layer and a slack-softmax output enforcing the common budget. Its 160-iteration L-BFGS fit distills feasible greedy actions. The convex pointwise improvement uses an explicit gradient-Lipschitz upper bound and checks its linear minimization gap rather than equating a

TABLE S.3
ALL RESOURCE RUNS AND THE MATCHED DEPLOYMENT ABLATION

d	Seed	NBO upper loss	Proposal only	Reference gap	Binding at 0
4	101	6.45e-04	1.19e-02	8.32e-08	0.812
4	202	4.88e-04	8.14e-03	7.40e-08	0.938
4	303	5.23e-04	6.31e-03	7.02e-08	0.938
8	101	4.29e-04	9.54e-02	9.79e-08	0.938
8	202	4.61e-04	3.28e-02	9.97e-08	1.000
8	303	2.03e-04	3.89e-02	9.75e-08	1.000
16	101	5.41e-04	6.73e-02	9.54e-08	1.000
16	202	5.67e-04	6.81e-02	9.63e-08	1.000
16	303	5.24e-04	6.65e-02	6.02e-08	1.000

Each row uses the stored 32 initial states and every one of the 64 terminal shock branches. The reference gap is the complete scenario-tree linear minimization gap at the tighter audit tolerance. Binding is the fraction of initial actions exhausting the common budget. All declared dimension-seed combinations are included.

stationary point with a maximum. All three seeds, 101, 202, and 303, are included at each dimension.

The independent scenario tree contains one root action, four next-date actions, and sixteen last-date actions for each tested initial state. Its objective sums all state and control costs and the 64 terminal costs with their probabilities. An independent backward adjoint computes the gradient. Reference optimization stops at a common-accuracy gap of 10^{-3} for timing and is separately continued to 10^{-7} for policy-cost auditing. Nonanticipativity is enforced by one action per information node, not one freely chosen action per complete shock path.

S.6.3. Reproduction Order

The repository README gives executable commands for the following sequence: generator, temporal, trace, utility, and game tests; stochastic grid refinements and re-solved adjustment-cost panels; the retained unsafeguarded neural pilot; all three safeguarded NDU runs; nine resource runs; independent frozen-policy and occupation analysis; computational-graph and checkpoint-replay tests; table generation; source/output manifest generation; and LaTeX builds. The table generator reads the deposited JSON records, so manually entered historical arrays cannot silently become current numerical evidence.

The source snapshot also retains the previous manuscripts, supplements, reports, reviewer diagnostics, and arrays unchanged. The preservation map locates their substantive topics in the new manuscript or this supplement. Retention is not endorsement of earlier incorrect claims. The development log records the standard-library module-name collision discovered during implementation, the corrected execution, and the budget difference of the pilot. Failed commands are not relabeled as successful scientific experiments.

S.6.4. What Each Numerical Error Means

For NDU, a reference value difference compares two specified finite approximations or evaluates a neural policy against a specified finite reference. The actor discrepancy reports consumption, preference adjustment, and portfolio separately because they

have different units. A feasible Bellman gain is measured in continuation-value units and uses an independently evaluated frozen continuation. It is a lower bound on the maximum profitable one-step deviation over a richer action set, not an upper bound on all deviations.

For resources, the full stochastic tree gives the evaluated policy cost. The reference tangent gap bounds its unknown optimum on that same tree, resulting in the interval in S.1. The held-out initial states are explicitly stored, so the bound is not extrapolated to all states. The Merton and Epstein–Zin policy errors compare learned homothetic coefficients with analytical solutions, while LQR errors compare structured learned policies with an independent Riccati solution. Temporal and game exploitability use separate best-response calculations against the actual continuation being evaluated. Trace standard errors concern a repeated-batch scalar loss and are not policy errors.

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